

1.3 Exit Note Solutions: Unit Conversions

Notation: $=$ joins values that are exactly equal; $\approx$ introduces a rounded value. Each conversion runs as one chain of unit factors and is evaluated once, so there is nothing to round in the middle.

1. Round to three significant figures.

(a) $\sqrt{3} = 1.7320508 \dots \approx \mathbf{1.73}$

(b) $\frac{100 \text{ m}}{9.58 \text{ s}} = 10.43841 \dots \approx \mathbf{10.4 \text{ m/s}}$

Three figures is also all the data allows: 9.58 s has three. The 100 m is a defined course length, not a measurement, so it does not limit the answer.

2. Scientific notation.

(a) $31,556,952 \text{ s} = 3.1556952 \times 10^7 \text{ s} \approx \mathbf{3.16 \times 10^7 \text{ s}}$

Move the decimal point 7 places left, so the exponent is +7. The familiar mnemonic “a year is $\pi \times 10^7$ seconds” gives 3.14×10^7 — good to two figures, wrong in the third.

(b) $1.06 \times 10^{-10} \text{ m} = \mathbf{0.000\,000\,000\,106 \text{ m}}$

A negative exponent of -10 moves the decimal point 10 places left: nine zeros after the point, then 106.

3. The Duomo di Milano, 108 m tall.

$$108 \text{ m} \times \frac{3.28 \text{ ft}}{1 \text{ m}} = 354.24 \text{ ft} \approx \mathbf{354 \text{ ft}}$$

The m cancel and ft remain. Three significant figures in, three out — converting a measurement never makes it more precise.

4. A Category 1 hurricane, 74 mi/h.

$$74 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{390,720}{3600} \frac{\text{ft}}{\text{s}} = 108.53 \dots \approx \mathbf{109 \text{ ft/s}}$$

Two factors, one evaluation. The mi cancel top-to-bottom and the h cancel bottom-to-top, leaving ft/s. Watch the second factor: to turn “per hour” into “per second” the hours must go on top, so the fraction is $\frac{1 \text{ h}}{3600 \text{ s}}$, not its reciprocal. If the answer comes out absurdly small, that factor was flipped.

5. Challenge: how fast is an elevator?

(a) Convert all four to m/s.

- Shanghai Tower: **20.5 m/s** (already in m/s)
- Taipei 101: $1010 \frac{\text{m}}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{1010}{60} \frac{\text{m}}{\text{s}} = 16.833 \dots \approx \mathbf{16.8 \text{ m/s}}$
- One World Trade Center: $2000 \frac{\text{ft}}{\text{min}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{2000}{3.28 \times 60} \frac{\text{m}}{\text{s}} = 10.162 \dots \approx \mathbf{10.2 \text{ m/s}}$
- Empire State (express): $1200 \frac{\text{ft}}{\text{min}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \times \frac{1 \text{ min}}{60 \text{ s}} = \frac{1200}{3.28 \times 60} \frac{\text{m}}{\text{s}} = 6.0975 \dots \approx \mathbf{6.10 \text{ m/s}}$

Write 6.10, not 6.1 — the trailing zero is the third significant figure and claiming it is the whole point of the instruction.

(b) Slowest to fastest: **Empire State (6.10) < One WTC (10.2) < Taipei 101 (16.8) < Shanghai Tower (20.5)**, all in m/s.

Yes, this ranking holds. The four speeds are separated by several m/s, and the shakiest figures here — “2000” and “1200” ft/min, whose trailing zeros are ambiguous and may carry only two significant figures — are still nowhere near each other. Even reading them as 2.0×10^3 and 1.2×10^3 ft/min, the gaps between the four are far larger than the uncertainty.

Compare with problem 4 on the 1.2 classwork, where the car at 26.8 m/s and the motorcycle at 27 m/s could not be told apart. Same question, opposite answer — and the difference is the size of the gap relative to the precision, not the number of digits the calculator shows. Give credit for any answer that checks the gaps against the precision, however it is worded.

(c) Time to rise 18 m, at the advertised speed:

$$t = \frac{d}{v}, \text{ so } \frac{18 \text{ m}}{20.5 \text{ m/s}} = 0.878 \dots \approx \mathbf{0.88 \text{ s}} \quad \frac{18 \text{ m}}{16.833 \text{ m/s}} = 1.069 \dots \approx \mathbf{1.1 \text{ s}}$$

$$\frac{18 \text{ m}}{10.162 \text{ m/s}} = 1.771 \dots \approx \mathbf{1.8 \text{ s}} \quad \frac{18 \text{ m}}{6.0975 \text{ m/s}} = 2.952 \dots \approx \mathbf{3.0 \text{ s}}$$

Two significant figures, because the 18 m has two. Carry the unrounded converted speeds into this division rather than the rounded ones.

Against 75 s on the stairs, even the slowest of these elevators is about **25 times faster**, and the Shanghai Tower machine about 85 times faster.

(d) **Why the times are all too small.** Each answer in (c) assumes the elevator is travelling at full speed for the entire 18 m. It is not: it starts at rest, speeds up, and must be back at rest by the top. Its *average* speed over the trip is therefore

less than its top speed, and the real ride takes longer than d/v predicts.

Why the Shanghai Tower is worst. The faster the elevator, the more distance it needs to get up to speed. At an acceleration passengers would tolerate — about 1 m/s^2 — reaching 20.5 m/s takes roughly 20 seconds and around 200 meters of shaft, and then as much again to stop. Our building gives it 18 m. That elevator would never come close to its advertised speed here; it would spend the whole ride speeding up and slowing down. The Empire State express at 6.10 m/s needs only about 6 seconds and 18 m, so it just barely reaches its top speed as it arrives.

The real lesson: an advertised top speed is not an average speed, and d/v only gives the travel time when the speed is constant the whole way. This is the same distinction as the Rome–Milan train, which is built for 300 km/h but averages about 190. We will make it precise when we reach acceleration.